MIL-F-163G(SH)
13 December 1985
SUPERSEDING
MIL-F-163F(SHIPS)
15 January 1968
(See 6.4)

MILITARY SPECIFICATION

FAUCETS, SINGLE AND DOUBLE, COMPRESSION AND SELF-CLOSING TYPE, NAVAL SHIPBOARD

This specification is approved for use within the Naval Sea Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers single and double, compression and self-closing faucets for plumbing installations on board ships of the U.S. Navy.

1.2 Classification. Faucets shall be of the following types and classes as specified (see 6.2).

Type I - Standard (bibb).
  Class a - Compression, plain-end.
  Class b - Compression, hose end.
  Class c - Self-closing.
Type II - Lavatory, self-closing, low spout.
  Class a - Fast-closing.
  Class b - Slow-closing.
Type III - Folding lavatory, self-closing.
Type IV - Elbow operated.
Type V - Lavatory, combination water supply and drain fixture (see 6.3.1.2).
Type VI - Combination supply fixture-swing spout for use on galley sinks.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, Naval Sea Systems Command, SEA 55Z3, Department of the Navy, Washington, DC 20362-5101 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 4510

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Type VII - Single supply fixture swing spout for use with steam kettles and urns.
Class a - Spout for overhead supply.
Class b - Spout for riser supply.
Type VIII - Supply fixture for pressed metal and shock mounted china lavatories.
Type IX - Lavatory, combination water supply and drain fixtures depression type (see 6.3.1.2).

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. The following specifications and standards form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DoDISS) and supplement thereto, cited in the solicitation.

SPECIFICATIONS

FEDERAL

QQ-C-320	- Chromium Plating (Electrodeposited).
QQ-N-290	- Nickel Plating (Electrodeposited).
PPP-B-566	- Boxes, Folding, Paperboard.
PPP-B-585	- Boxes, Wood, Wirebound.
PPP-B-591	- Boxes, Shipping, Fiberboard, Wood-Cleated.
PPP-B-601	- Boxes, Wood, Cleated-Plywood.
PPP-B-621	- Boxes, Wood, Nailed and Lock-Corner.
PPP-B-636	- Box, Shipping, Fiberboard.
PPP-B-676	- Boxes, Setup.

MILITARY

MIL-P-116	- Preservation, Methods of.

STANDARDS

FEDERAL

FED-STD-H28	- Screw-Thread Standards for Federal Services.

MILITARY

MIL-STD-105	- Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129	- Marking for Shipment and Storage.

2.1.2 Government drawings. The following Government drawings form a part of this specification to the extent specified herein. Unless otherwise specified, the issues shall be those in effect on the date of the solicitation.

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DRAWINGS

NAVAL SEA SYSTEM COMMAND (NAVSEA)

NAVSEA 803-5959206 - Faucet, Metering.
NAVSHIPS 810-1385771 - Lavatory, Vitreous China, and Shock Mounts.
NAVSHIPS 810-1385890 - Lavatory, Pressed Metal.

(Copies of specifications, standards, and drawings required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting activity.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of the documents which are DoD adopted shall be those listed in the issue of the DoDISS specified in the solicitation. Unless otherwise specified, the issues of documents not listed in the DoDISS shall be the issue of the non-government documents which is current on the date of the solicitation.

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

B 62 - Standard Specification for Composition Bronze or Ounce Metal Casting. (DoD adopted)
B 584 - Standard Specification for Copper Alloy Sand Casting for General Applications. (DoD adopted)

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

UNIFORM CLASSIFICATION COMMITTEE AGENT

Uniform Freight Classification Ratings, Rules and Regulations

(Application for copies should be addressed to the Uniform Classification Committee Agent, Tariff Publication Officer, Room 1106, 222 South Riverside Plaza, Chicago, IL 60606.)

(Nongovernment standards and other publications are normally available from the organizations which prepare or which distribute the documents. These documents also may be available in or through libraries or other informational services.)

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein (except for associated detail specifications, specification sheets or MS standards), the text of this specification shall take precedence. Nothing in this specification, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

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3. REQUIREMENTS

3.1 General. Requirements for faucet types I through VIII shall be in accordance with section 3 of this specification. Requirements for faucet type IX shall also be in accordance with Drawing 803-5959206.

3.2 Material.

3.2.1 Faucet bodies shall be made of bronze conforming to ASTM B 62 or ASTM B 584.

3.2.2 The parts shall be made of the materials specified in table I.

TABLE I. Material for parts.

Part	Material
Handles	Cast or forged brass or bronze.
Bonnets	Commercial brass or bronze.
Machine screws	Commercial brass.
Seat washers	Rubber or fiber material capable of withstanding the hydrostatic test specified in 4.4.1 and suitable for water temperature from 30 to 190 degrees Fahrenheit (°F) without deterioration of material.
Springs for self-closing faucets	Corrosion-resisting steel, phosphor bronze or nickel copper.
Handwheel and nut	Brass or bronze.

3.2.3 Recovered materials. Unless otherwise specified herein, all equipment, material, and articles incorporated in the products covered by this specification shall be new and may be fabricated using materials produced from recovered materials to the maximum extent practicable without jeopardizing the intended use. The term "recovered materials" means materials which have been collected or recovered from solid waste and reprocessed to become a source of raw materials, as opposed to virgin raw materials. None of the above shall be interpreted to mean that the use of used or rebuilt products is allowed under this specification unless otherwise specifically specified.

3.3 Handles. Handles shall be four ball, four arm, star, T, loose-key T, or lever type as specified (see 6.2), except that handles for type II, class b, faucets shall be button type. Handwheels may be provided for type VII faucets.

3.4 Threads. Threads for threaded parts shall conform to the applicable requirements of FED-STD-H28.

3.5 Finish. Faucets (all exposed metal trim) shall be nickel chromium plated.

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3.5.1 Prior to plating, faucets shall be thoroughly cleaned of sand and scale by the acid-bath process (or a process satisfactory to the command or agency concerned) and shall have all fins and roughness removed, and polished.

3.5.2 Nickel chromium plating shall be applied to surfaces normally visible after installation including exposed bolts, nuts, screws, or other fastenings.

3.5.2.1 Nickel plating on copper base alloys shall be class 1, type VII(QC) in accordance with QQ-N-290.

3.5.2.2 Chromium plating, over nickel, shall be class 1, type I in accordance with QQ-C-320.

3.6 Marking.

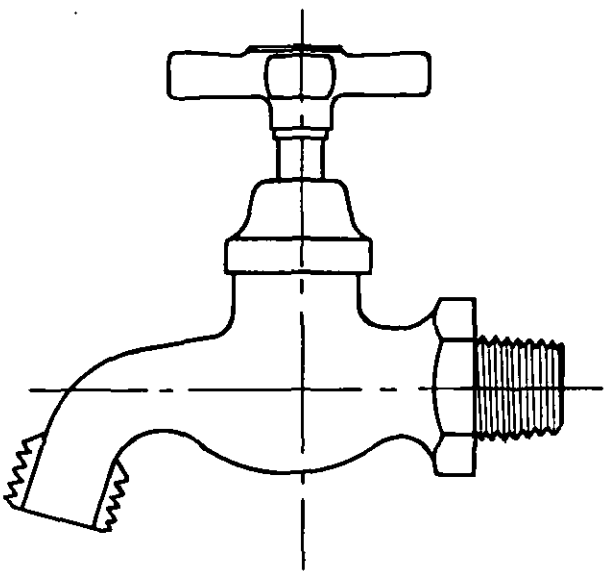
3.6.1 All handles shall be marked "hot" or "cold" as specified (see 6.2). Substitution of "H" or "C" is permissible for "hot" or "cold", respectively.

3.6.2 Each faucet shall be legibly stamped prior to plating, with the name or identification mark of the manufacturer.

3.7 Hydrostatic pressure. Each sample faucet shall withstand the hydrostatic pressure tests specified in 4.4.1.

3.8 Type I, standard (bibb).

3.8.1 Type I faucets shall be in accordance with figure 1. They shall have hexagon shoulders and a male threaded inlet, and shall be of the sizes and weights in table II as specified (see 6.2).



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FIGURE 1. Faucet, type I, classes a, b, and c.

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TABLE II. Sizes and weight of type I faucets.

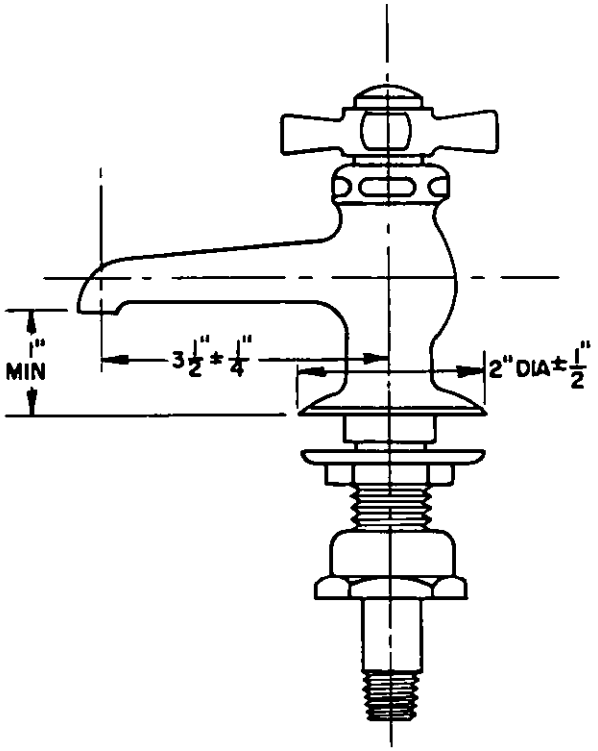
Nominal pipe size (nps)	Minimum weight
(Inches)	(Pounds)
1/2	3/4
3/4	7/8
1	1
1-1/4	1-1/8

3.8.2 Class a, b, and c. Class a faucets shall be plain. Class b shall have 3/4 inch hose thread. Class c shall be the same as class a except that they shall be self-closing.

3.9 Type II lavatory, self-closing, low spout.

3.9.1 Type II faucets shall be provided with a shank, fitted with a metal washer and locknut and with a 1/4-inch nps tailpiece. The shank of the faucet and one end of the tailpiece shall be arranged as a ground joint union; the other end of the tailpiece shall be provided with outside threads 1/4-inch nps. The minimum weight of each faucet, including tailpiece shall be 1-1/2 pounds plus or minus 5 percent.

3.9.2 Class a. Class a lavatory faucets shall be in accordance with figure 2. They shall operate on a cam, ball or roller bearings. Faucets with renewable operating unit will be considered. Overall dimensions shall be as specified (see 6.2).



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FIGURE 2. Faucet, type II, class a.

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3.9.3 Class b. Class b lavatory faucets shall be in accordance with figure 3. They shall be fitted with an adjusting device, that cannot be tampered with, for regulating the closing from 3 to 15 seconds. Overall dimensions shall be as specified (see 6.2). Dimensions shown on figure 3 are approximate. Shank design shall be as specified (see 6.2).

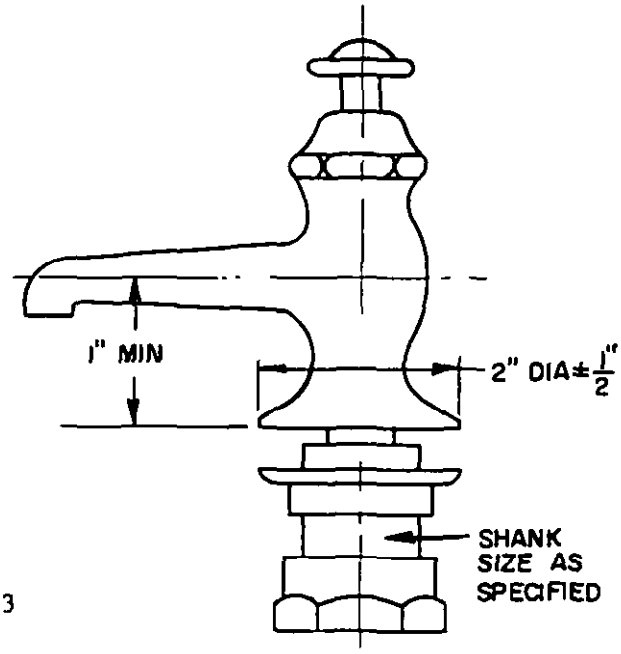


FIGURE 3. Faucet, type II, class b.

3.10 Type III, folding lavatory self-closing.

3.10.1 Type III faucets shall be in accordance with figure 4. They shall be of the cam, ball or roller bearing self-closing type. The body shall be designed for back mounting. Overall dimensions and position of inlet (side or bottom) shall be as specified (see 6.2). Dimensions shown on figure 4 are approximate.

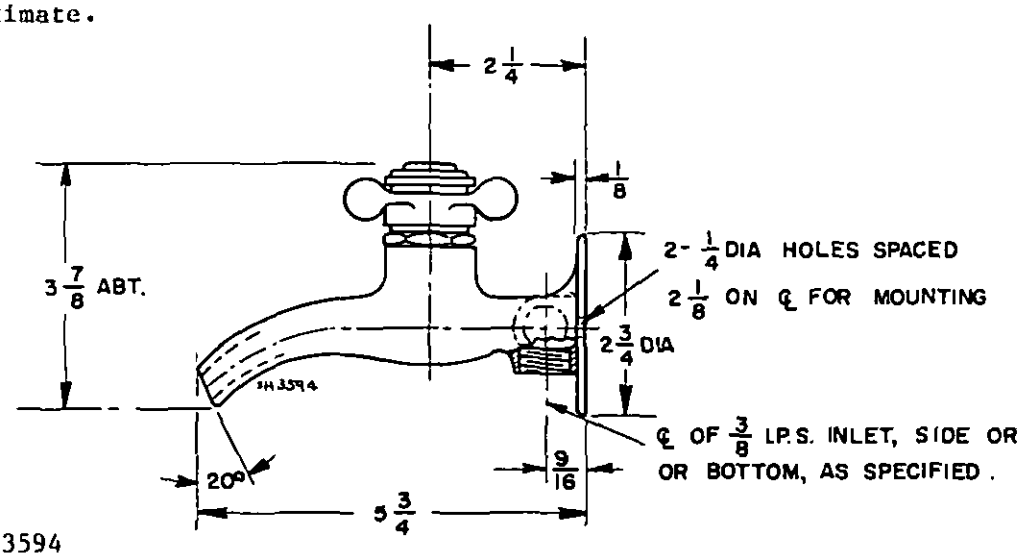
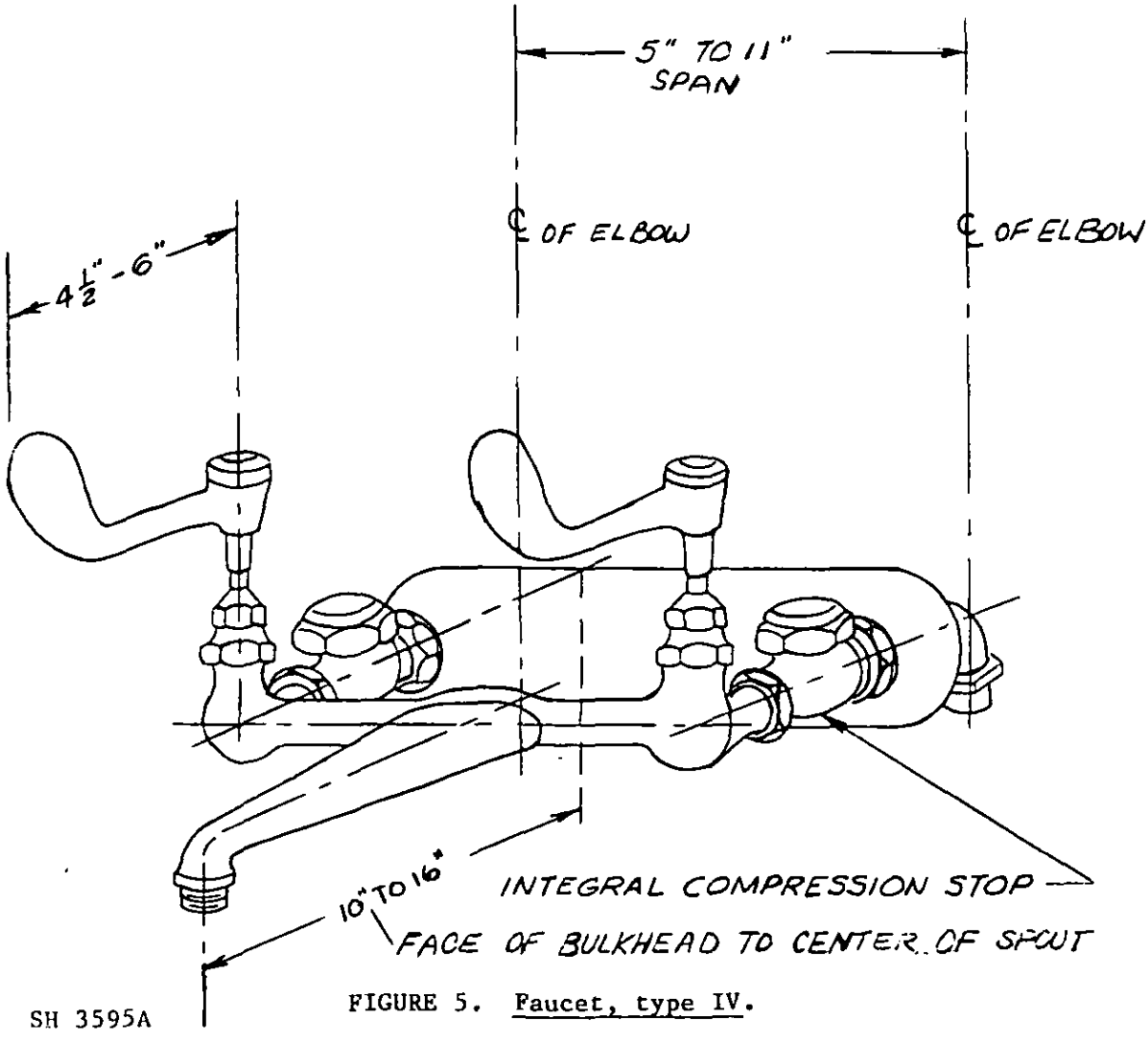


FIGURE 4. Faucet, type III.

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3.11 Type IV, elbow operated.

3.11.1 Type IV faucets shall be in accordance with figure 5 except that goose neck spout is acceptable. They shall have a double faucet with a rigid spout with or without buttons, renewable seats and sleeves or renewable units and integral stops in shanks with union inlets. Escutcheon plates shall be provided for the fixtures. Overall dimensions shall be as specified (see 6.2). Spray shall be provided when specified (see 6.2). Dimensions shown on figure 5 are approximate.



SH 3595A

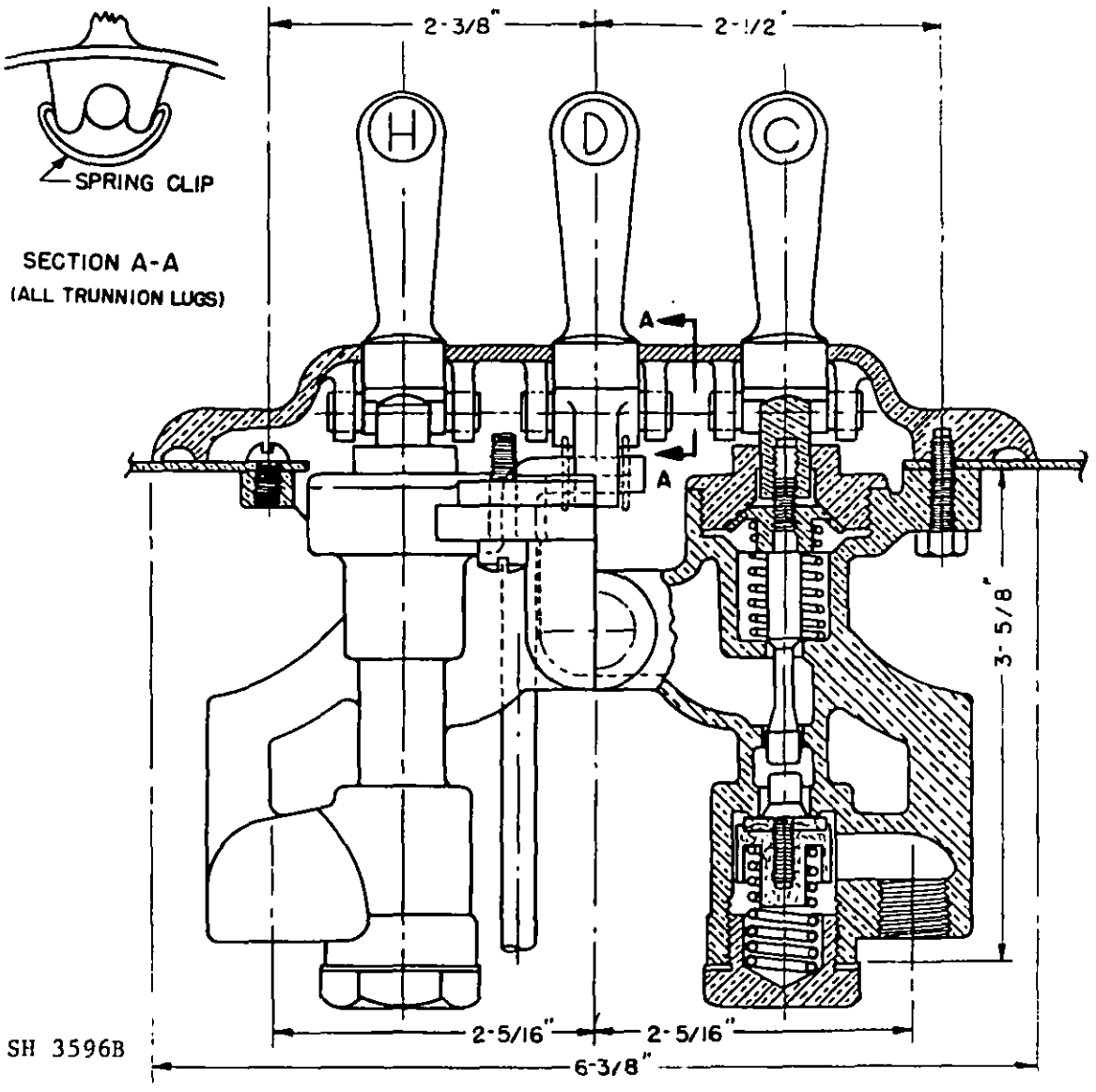
FIGURE 5. Faucet, type IV.

3.12 Type V, lavatory, combination water supply and drain fixtures (see 6.3.1.2).

3.12.1 Type V faucets shall conform to figure 6.

3.12.2 Trunnion lugs shall be designed so that the trunnions will be held in place by spring clips or other device.

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Note: See 6.3.1.2 for repair or replacement of existing faucets only. Not for new construction.

FIGURE 6. Type V faucet.

3.13 Type VI, combination supply fixture-swing spout for use on galley sinks.

3.13.1 Type VI fixtures shall be in accordance with figure 7. They shall have manifolded hot and cold compression type faucets, discharging through a common swing spout, 8 inches or 14 inches minimum length, as specified (see 6.2). The swing joint shall be of rugged design and packed with material resistant to water and leakage at temperatures of 30 to 200°F. The faucet shall have renewable seats, sleeves, or renewable units and stops in the body and shanks with union inlet connection, with or without escuthcheon plates as specified in the contract or order. Overall dimensions shall be as shown on figure 7.

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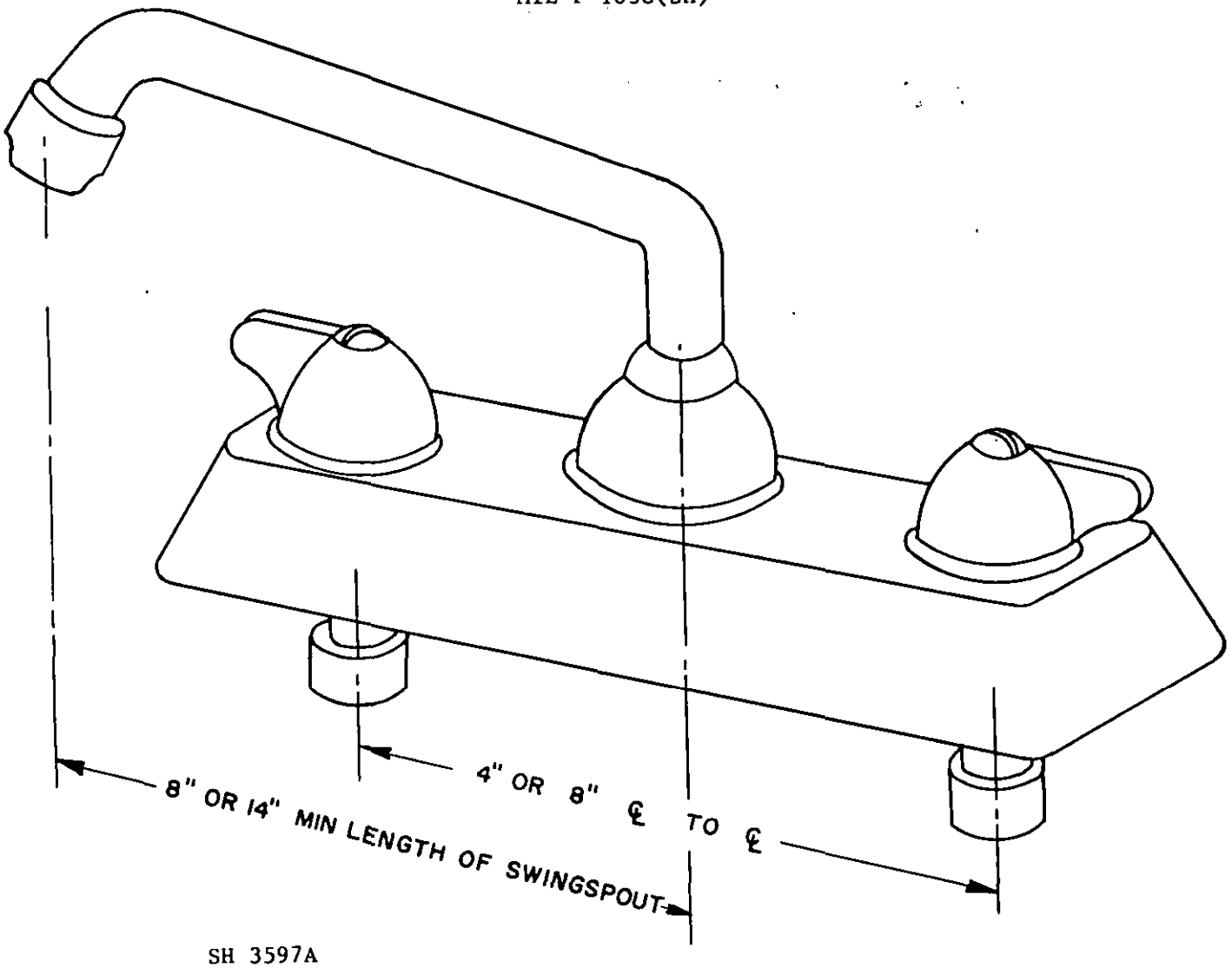


FIGURE 7. Faucet type VI.

3.14 Type VII, single supply fixture swing spouts for use with steam kettles and urns.

3.14.1 Type VII fixtures shall be in accordance with figure 8. They shall consist of a single globe type valve with nps threaded inlet and outlet and an adapter designed to receive a swing spout. The swing joint shall be ruggedly constructed and fitted with packing material resistant to leakage and water at temperatures of 30 to 200°F.

3.14.2 Class a and b spouts. Spouts shall be class a or b as specified (see 6.2) and may be of cast or extruded construction. Class a or b spouts shall be designed for installation on the valve. Overall dimensions shall be as shown on figure 8.

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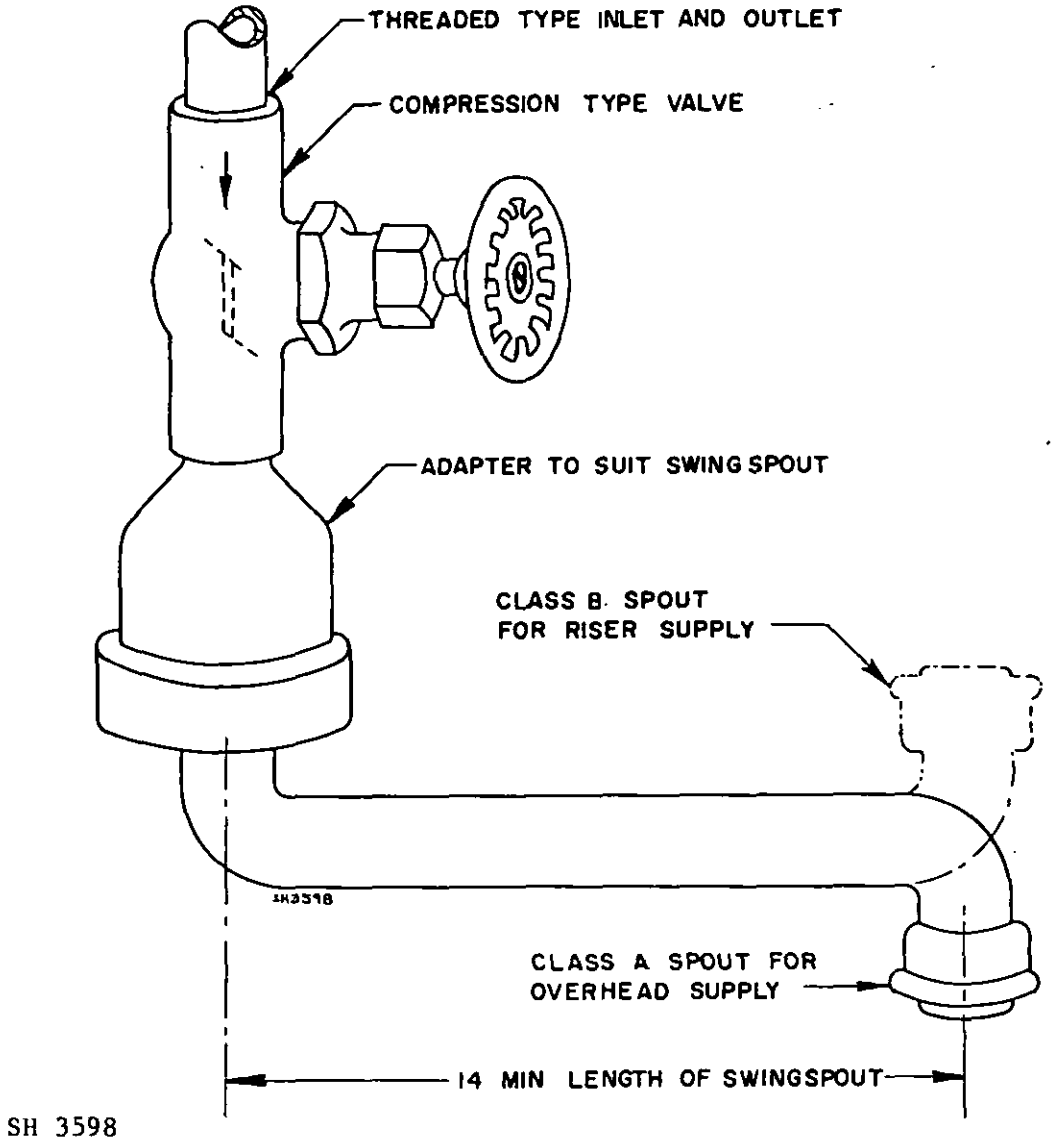
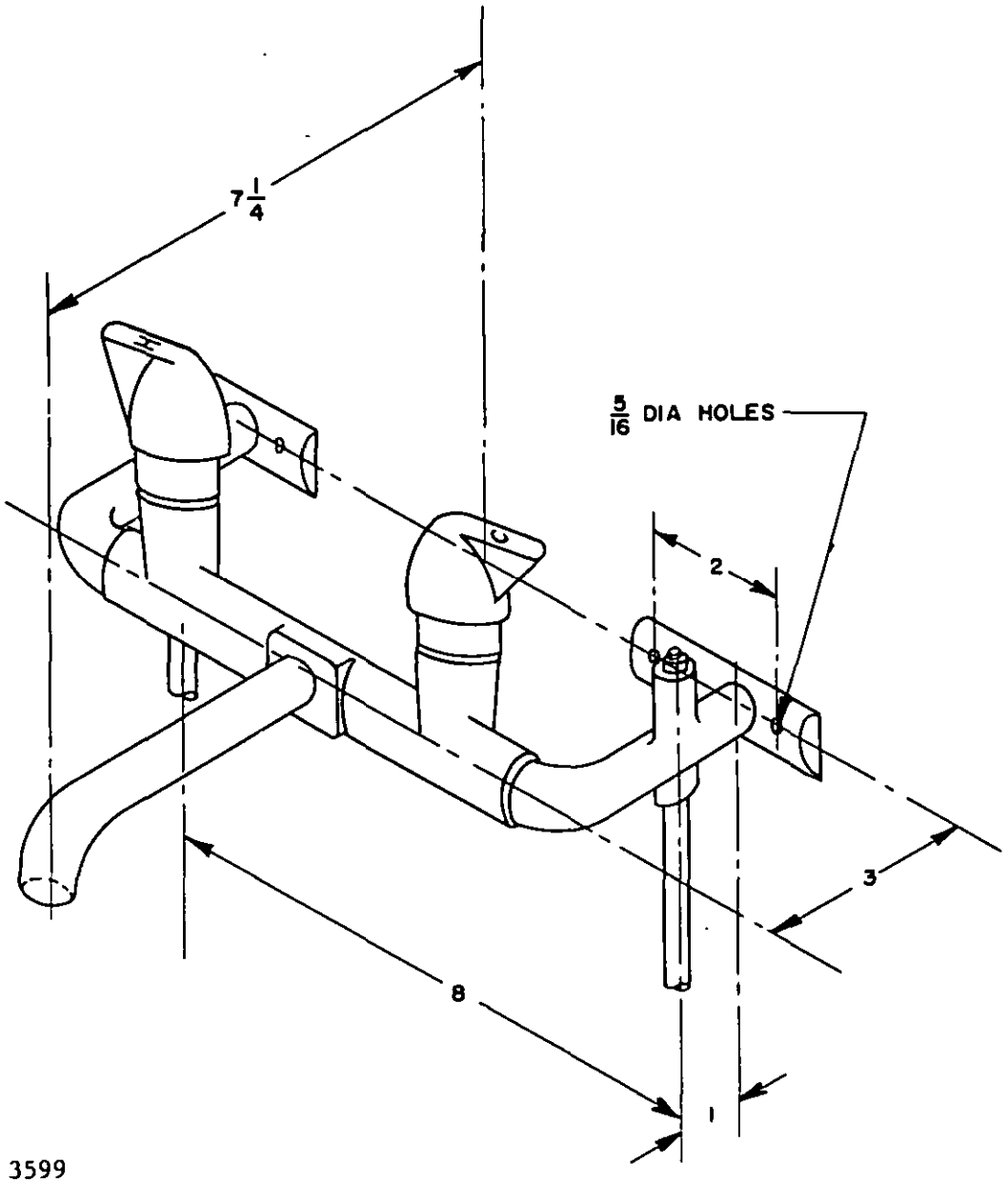


FIGURE 8. Faucet, type VII, classes a and b.

3.15 Type VIII, supply fixture for pressed metal and shock mounted china, lavatories.

3.15.1 Type VIII fixtures shall be in accordance with figure 9. They shall consist of a manifold incorporating hot and cold water nps female threaded inlet connections to permit connection from top or bottom of the manifold and common outlet (through direct compression type or inverted compression type) faucets. Unless otherwise specified in the contract or order, the connections shall be 1/4 inch nps. The faucets shall be of the self-closing type operating on a cam, ball or roller bearing. The faucet may seat with or against the water supply pressure. Faucets with renewable operating unit will be acceptable. Drawing 810-1385771 and Drawing 810-1385890 showing pressed metal lavatories and faucets with renewable operating unit will be acceptable. Overall dimensions shall be as shown on figure 9.

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SH 3599

FIGURE 9. Faucet, type VIII.

3.16 Workmanship. Faucets shall be true to form and free from all defects or blemishes affecting the appearance or serviceability.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

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4.1.1 Responsibility for compliance. All items must meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of assuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling in quality conformance does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material.

4.2 Quality conformance inspection. Quality conformance inspection shall consist of the examination of 4.3 and the tests specified in 4.4.

4.2.1 Lot. All faucets of the same type, class and size offered for delivery at one time shall be considered a lot for purposes of examination and hydrostatic test.

4.2.2 Sampling for examination and hydrostatic pressure test. Faucets from each lot (see 4.2.1) shall be selected in accordance with inspection level I of MIL-STD-105 for examination and hydrostatic pressure test. The acceptable quality level (AQL) shall be 1.5 percent defective.

4.3 Visual and dimensional examination. Sample valves shall be visually and dimensionally examined to determine conformance with the specified design, inclusion of all required parts, proper assembly, the existence of proper marking and dimension, specified on the applicable specification, standards and drawings.

4.4 Tests.

4.4.1 Hydrostatic test. Samples selected in accordance with 4.2.2 shall be tested under a hydrostatic pressure of 100 pounds per square inch (lb/in^2). Faucets which do not operate satisfactorily or which show leakage shall not be offered for delivery.

4.4.2 Metering tests for type IX faucets only. Samples selected in accordance with 4.2.2 shall be tested for metering requirements specified on Drawing 803-5959206 by timing the open to close cycle time at 20 lb/in^2 inlet pressure and at 100 lb/in^2 inlet pressure. Cycle time at 20 lb/in^2 shall not exceed 20 seconds and cycle time at 100 lb/in^2 shall not be less than 3 seconds. Each sample faucet shall be cycled a minimum of 10 times under the pressures noted. Failure to operate within the specified times or to be tight after each cycle shall be cause for rejection.

4.5 Inspection of packaging. Sample packages and packs, and the inspection of the preservation-packaging, packing and marking for shipment and storage shall be in accordance with the requirements of section 5 and the documents specified therein.

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5. PACKAGING

(The packaging requirements specified herein apply only for direct Government acquisition.)

5.1 Domestic shipment and storage or overseas shipment. The requirements, and levels of packaging, packing and marking for shipment shall be as specified by the contracting agency (see 6.2).

5.1.1 The following provides various levels for protection during domestic shipment and storage or overseas shipment, which may be required when acquisition is made.

5.2 Packaging.

5.2.1 Level A. The faucets shall be packaged in accordance with method III of MIL-P-116. Unit and intermediate containers shall conform to PPP-B-566, PPP-B-676, or PPP-B-636. Quantity per unit and intermediate container shall be as specified by the command or agency concerned.

5.2.1.1 Level C. Packaging which may be the contractor's commercial practice shall be sufficient to afford adequate protection against corrosion, deterioration and physical damage during shipment from the supply source to the using activity and until early installation.

5.2.1.2 Packing.

5.2.1.2.1 Level A. The faucets, packaged as specified shall be packed in containers conforming to any one of the following specifications at the option of the contractor:

Specification	Type or class
PPP-B-591	Class II
PPP-B-601	Overseas type
PPP-B-621	Class 2
PPP-B-585	Class 3
PPP-B-636	Type SF, Weather resistant class

The gross weight of wood type boxes shall not exceed 200 pounds; fiberboard boxes shall not exceed the weight limitations of the box specification. Closures shall be in accordance with the applicable box specification and the appendix thereto.

5.2.1.2.2 Level B. The faucets, packaged as specified shall be packed in containers conforming to any one of the following specifications at the option of the contractor:

Specification	Type or class
PPP-B-591	Class I
PPP-B-601	Domestic
PPP-B-621	Class 1
PPP-B-585	Class 1
PPP-B-636	Type SF, Domestic class

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The gross weight of wood type boxes shall not exceed 200 pounds; fiberboard boxes shall not exceed the weight limitations of the applicable box specification. Closure shall be in accordance with applicable box specification and the appendix thereto.

5.2.1.2.3 Level C. Packing shall be accomplished in a manner which shall ensure acceptance by common carrier at the lowest rate and shall afford protection against physical or mechanical damage during direct shipment from the supply source to the using activity for early installation. The shipping containers or method of packing shall conform to the Uniform Freight Classification Rules or other carrier regulations as applicable to the mode of transportation and may conform to the contractor's commercial practice.

5.3 Marking. In addition to any special marking required in the contract or order, marking of the packages and shipping containers shall be in accordance with MIL-STD-129.

6. NOTES

6.1 Intended use. Single and double, compression and self-closing faucets covered by this specification are intended for plumbing installations on board ships of the U.S. Navy.

6.2 Ordering data. Acquisition documents should specify the following:

(a) Title, number, and date of this specification.
(b) Type and class required (see 1.2).
(c) Type of handle required (see 3.3).
(d) Marking required for handles (see 3.6.1).
(e) Size and weight of type I faucet required (see 3.8.1).
(f) Overall dimensions for type II, class a faucets (see 3.9.2).
(g) Overall dimensions and shank design for type II, class b faucets (see 3.9.3).
(h) Overall dimensions and position of inlet (side or bottom) for type III faucets (see 3.10.1).
(i) Overall dimensions and whether spray is required for type IV faucets (see 3.11.1).
(j) Length of spout for type VI fixtures (see 3.13.1).
(k) Whether type VII spouts are to be class a or b (see 3.14.2).
(l) Preservation, packaging, packing, and marking requirements if other than those specified (see 5.1, 5.2, and 5.3).

6.3 Provisioning. Provisioning Technical Documentation (PTD), spare parts, and repair parts should be furnished as specified in the contract.

6.3.1 When ordering spare parts or repair parts for the equipment covered by this specification, the contract should state that such spare parts and repair parts should meet the same requirements and quality assurance provisions as the parts used in the manufacture of the equipment. Packaging for such parts should also be specified.

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6.3.1.1 On board repair parts. On board repair parts shall be furnished for shipboard installations in accordance with table III for each type valve installed, with the exception of type IX valves which shall be in accordance with Drawing 803-5959206.

TABLE III. On board repair parts.

Item	Quantity
Complete faucet	1 for each ten faucets installed.
Rubber discs, packing and seals	100 percent for each faucet.
Springs and screws	1 set for each five faucets installed.
Handles or handwheels	1 for each ten faucets installed.

6.3.1.2 Type V faucets and repair parts for type V faucets are for replacement or repair of existing lavatories only. For lavatories in accordance with Drawing 805-1623970, type IX, combination water supply and drain fixture shall be supplied.

6.4 Changes from previous issue. Asterisks are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

Preparing activity:
Navy - SH
(Project 4510-N221)

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-F-163G(SH)	Faucets, Single And Double, Compression And Self-Closing Type, Naval Shipboard

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ USER
	☐ MANUFACTURER
	☐ OTHER *(Specify)*:

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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PREVIOUS EDITION IS OBSOLETE.

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